Q1. What are Modules? Explain the need for Modules.

Ans:

A module in Python is a file that contains Python code such as functions, variables and classes. Modules help us organize code and reuse it in different programs.

Need of Modules:

1. Code Reusability: We can use the same code in different programs without writing it again.
2. Code Organization: Modules help us organize large programs into smaller files.
3. Easy Maintenance: It becomes easier to update, manage and debug code.
4. Avoid Rewriting Code: We can use already available functions from different modules.
5. Better Readability: Modules make programs easier to understand.

Example:

Python provides a built-in math module.

```
import math


print(math.sqrt(25))
```

Output:

5.0

Here, math is a module and sqrt() is a function used to calculate the square root.

Q2. State the Different Ways of Importing a Module.

Ans:

In Python, there are different ways to import a module into a program.

1. Import the entire module

We use the import keyword to import the complete module.

```
import math


print(math.sqrt(25))
```

2. Import a specific function

We use from and import to import a particular function from a module.

```
from math import sqrt
```

print(sqrt(25))

3. Import multiple functions

We can import multiple functions from the same module.

from math import sqrt, factorial

print(sqrt(25))

print(factorial(5))

4. Import a module with an alias

We use the as keyword to give a shorter name to a module.

import math as m

print(m.sqrt(25))

5. Import all functions from a module

We use * to import all names from a module.

from math import *

print(sqrt(25))

print(factorial(5))

Difference between Importing Methods:

Method	**Meaning**
import math	Imports the complete module.
from math import sqrt	Imports a specific function.
from math import sqrt, factorial	Imports multiple functions.
import math as m	Imports a module with an alias.
from math import *	Imports all available names from the module.

Remember:

- import = Import a complete module.
- from ... import = Import specific functions.
- as = Give an alternative name (alias).
- * = Import all names.